

LANZHOU ZHONGCHUAN, China

WMO: 525331

Lat: 36.515N

Lon: 103.620E

Elev: 1947

StdP: 80.02

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-17.0	-24.2	0.5	-13.4	-23.0	0.6	-11.8	7.4	-3.6	6.5	-5.0	2.0	350	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.1	15.8	28.8	15.6	27.0	14.9	18.6	25.5	17.8	24.2	17.0	23.1	3.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	15.4	20.6	15.9	14.4	20.2	15.0	13.5	19.6	62.0	25.7	59.0	24.8	56.0	22.6	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.8	5.6	-21.7	32.8	2.9	1.5	-23.8	33.9	-25.5	34.8	-27.1	35.6	-29.2	36.7		
			-22.2	19.8	2.7	0.9	-24.2	20.4	-25.8	20.9	-27.3	21.4	-29.3	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-8.0	-3.8	3.1	9.3	13.8	18.0	20.1	19.2	14.0	8.0	0.6	-6.4	
	DBStd	10.22	3.47	4.25	4.14	3.79	3.26	2.52	2.62	2.80	2.58	3.20	3.60	3.03	
	HDD10.0	2089	558	387	214	55	7	0	0	0	2	76	282	508	
	HDD18.3	4139	817	621	471	270	141	35	12	24	132	319	532	766	
	CDD10.0	1137	0	0	2	35	126	239	313	284	122	16	0	0	
	CDD18.3	145	0	0	0	0	2	24	67	49	2	0	0	0	
	CDH23.3	1652	0	0	1	18	94	336	708	466	28	2	0	0	
	CDH26.7	409	0	0	0	2	9	62	220	116	1	0	0	0	
Wind	WSAvg	2.4	2.1	2.5	2.9	3.1	2.9	2.5	2.4	2.4	2.1	2.2	2.1	2.0	
Precipitation	PrecAvg	293	2	3	7	15	35	43	62	60	43	19	3	1	
	PrecMax	372	7	10	18	63	81	84	105	94	70	45	7	3	
	PrecMin	221	0	1	0	1	7	17	10	26	14	2	0	0	
	PrecStd	43	2	2	5	12	19	19	27	19	16	9	2	1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	13.1	20.8	25.9	27.9	30.1	33.0	31.2	26.0	21.8	14.2	7.1	
		MCWB	-0.9	2.4	7.2	10.1	12.0	14.8	17.0	16.9	14.6	10.6	4.8	-0.4	
	2%	DB	4.2	9.1	16.9	22.2	25.9	28.8	30.8	29.2	23.9	19.1	12.1	4.8	
		MCWB	-2.4	1.0	5.1	8.2	11.2	13.9	16.5	16.4	13.2	9.8	3.6	-1.5	
	5%	DB	1.9	7.0	14.1	20.1	24.0	27.0	29.1	27.9	22.1	16.9	10.0	2.8	
		MCWB	-3.7	-0.2	3.9	7.6	10.7	13.5	16.1	16.0	12.9	8.9	3.0	-2.6	
	10%	DB	-0.2	4.2	12.0	18.0	22.1	25.2	27.2	26.0	20.2	14.9	7.9	0.9	
		MCWB	-4.9	-1.6	2.8	6.8	9.9	12.9	15.9	15.5	12.6	8.0	1.9	-3.7	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	3.4	7.8	11.3	14.4	17.7	19.9	19.7	16.5	12.8	5.9	0.1	
		MCDB	6.9	11.6	17.6	20.1	22.6	23.2	27.5	26.3	21.9	18.2	12.0	5.8	
	2%	WB	-2.4	1.2	5.6	9.8	13.1	16.6	18.9	18.6	15.4	11.2	4.6	-1.3	
		MCDB	4.0	8.4	14.8	19.3	21.0	22.6	26.2	24.9	20.2	16.3	10.5	3.9	
	5%	WB	-3.8	-0.1	4.4	8.8	12.0	15.6	18.0	17.9	14.4	10.0	3.5	-2.5	
		MCDB	1.3	6.3	13.0	18.0	20.0	21.9	25.4	23.9	19.1	14.9	9.0	2.3	
	10%	WB	-4.9	-1.4	3.2	7.6	11.1	14.6	17.1	17.1	13.6	8.9	2.2	-3.6	
		MCDB	-0.7	4.1	11.1	15.8	19.3	21.6	23.9	22.7	18.5	13.6	7.1	0.4	
Mean Daily Temperature Range	5% DB	MDBR	14.2	13.9	13.8	14.0	13.2	12.7	12.3	11.4	10.9	11.8	12.8	14.1	
		MCDBR	18.1	18.1	18.3	18.5	17.2	16.5	16.3	14.9	14.9	15.4	15.7	16.8	
		MCWBR	12.6	11.3	9.5	8.5	6.7	5.4	5.2	5.0	6.2	7.6	8.7	11.3	
	5% WB	MCDBR	17.1	17.1	16.7	15.7	14.1	11.5	12.5	10.9	11.0	12.9	14.6	15.8	
		MCWBR	12.1	10.8	8.8	7.5	6.0	4.5	4.7	4.5	5.4	6.9	8.4	10.8	
Clear-Sky Solar Irradiance	taub	0.328	0.385	0.447	0.486	0.487	0.477	0.461	0.474	0.449	0.411	0.345	0.315		
	taud	2.149	1.989	1.855	1.805	1.875	1.963	2.084	2.022	2.036	2.048	2.154	2.209		
	Ebn at Noon	841	827	808	801	807	814	824	804	800	789	813	832		
	Edh at Noon	113	151	190	211	200	183	161	168	157	141	112	100		
All-Sky Solar Radiation	RadAvg	2.95	3.78	4.82	5.63	6.07	6.01	5.82	5.34	4.19	3.79	3.17	2.81		
	RadStd	0.27	0.23	0.28	0.33	0.35	0.32	0.50	0.34	0.31	0.33	0.21	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page